

Learning Objective 2.3: I can describe the radiation mechanism, pattern, and typical use cases for patch, slot, and horn antennas.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate). Take $c = 3 \times 10^8$ m/s and $\eta_0 = 377 \Omega$.

1. **Sizing a patch.** You are designing a rectangular microstrip patch to resonate at $f_r = 3.5$ GHz on Rogers RO4003C: $\epsilon_r = 3.55$, substrate thickness $h = 0.813$ mm. The transmission-line design set is

$$W = \frac{c}{2f_r} \sqrt{\frac{2}{\epsilon_r + 1}} \quad \epsilon_{\text{eff}} = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \left(1 + \frac{12h}{W}\right)^{-1/2}$$

$$\frac{\Delta L}{h} = 0.412 \frac{(\epsilon_{\text{eff}} + 0.3)(W/h + 0.264)}{(\epsilon_{\text{eff}} - 0.258)(W/h + 0.8)} \quad L = \frac{c}{2f_r \sqrt{\epsilon_{\text{eff}}}} - 2\Delta L$$

- (a) Find the patch width W in millimetres.

$$W = \boxed{}$$

- (b) Find the effective permittivity ϵ_{eff} .

$$\epsilon_{\text{eff}} = \boxed{}$$

- (c) Find the edge extension ΔL and the physical length L .

$$\Delta L = \boxed{}$$

$$L = \boxed{}$$

- (d) The half-wavelength in the dielectric is $\lambda_d/2 = 23.3$ mm, yet you etched $L = 22.6$ mm. Explain in one or two sentences what physical effect forces the metal to be cut short, and state which way the resonant frequency moves if you etch the full 23.3 mm instead.

2. **The substrate trade.** A patch is a high- Q cavity, and its impedance bandwidth for $\text{VSWR} \leq 2$ follows the closed form

$$\text{BW} \approx 3.77 \frac{\varepsilon_r - 1}{\varepsilon_r^2} \frac{h}{\lambda_0} \frac{W}{L}$$

Continue with the 3.5 GHz design of Question 1 ($W = 28.41$ mm, $L = 22.55$ mm, $h = 0.813$ mm, $\varepsilon_r = 3.55$).

- (a) Estimate the fractional bandwidth of the RO4003C design, as a percentage.

BW =

- (b) Rebuild the same 3.5 GHz patch on $\varepsilon_r = 10.2$ at the same h . Its design works out to $W = 18.11$ mm, $L = 13.35$ mm. Find the new bandwidth, and the factor by which the patch *area* shrank.

BW =

area ∇ .

- (c) Read the bandwidth formula: which single substrate parameter would you change to *buy* bandwidth without changing ε_r , and name one penalty that comes with it.
- (d) You are given $8 \text{ mm} \times 8 \text{ mm}$ of board area for a 3.5 GHz element and told the link needs 2% bandwidth. State whether a single patch on any of these substrates can do the job, and what you would tell the systems lead.

3. **Babinet and the slot.** A narrow half-wave slot is cut in a large, thin conducting sheet and fed across its middle.

- (a) State Babinet's complementarity relation between a slot and its complementary dipole, and say in one sentence what "complementary" means physically.

- (b) The complementary dipole is trimmed to resonance, where it presents $73 \, \Omega$ real. Find the slot's input impedance.

$Z_{\text{slot}} =$

- (c) Now take the untrimmed $\lambda/2$ dipole, $Z = 73 + j42.5 \, \Omega$. Find the slot impedance and comment on the sign of its reactance.

$Z_{\text{slot}} =$

- (d) A horizontal slot is cut in the vertical fin of an aircraft. State the polarization of the radiated field and justify it in one sentence, then name one reason a designer chooses a slot here rather than a monopole.

4. **An X-band horn.** A pyramidal standard-gain horn has a rectangular aperture $16.5 \text{ cm} \times 12.2 \text{ cm}$ and is used at 9.4 GHz. Aperture antennas obey $G = \eta_{ap} 4\pi A/\lambda^2$; take the optimum-horn value $\eta_{ap} = 0.5$ unless told otherwise.

(a) Find the gain, in dBi.

$$G = \boxed{}$$

(b) How far away must a receiving antenna sit for this horn to be in its far field?

$$r = \boxed{}$$

(c) A shorter, more sharply flared horn of the same aperture has enough edge phase error to drop η_{ap} to 0.35. What gain does it deliver, and how much was lost?

$$G = \boxed{}$$

$$\text{loss} = \boxed{}$$

(d) Explain in two or three sentences why enlarging a horn's aperture at fixed flare length eventually stops increasing its gain, and what the phrase "optimum horn" names.

5. **Mechanism and selection.** Answer in words. Full sentences.

- (a) A patch is mostly a flat sheet of copper over a ground plane, and a sheet of copper is a terrible radiator. Explain what actually radiates, and why the two contributions add rather than cancel in the broadside direction.

- (b) The two radiating slots sit only about a third of a free-space wavelength apart. Say what that implies for the E-plane beamwidth, and why the patch's directivity therefore lands near 6 dBi rather than up with the horns.

- (c) Match each application to the best of patch, slot, or horn, with a one-line justification: **(i)** the gain reference in an anechoic chamber; **(ii)** a telemetry antenna on a supersonic missile skin; **(iii)** the element of a printed phased array for a 10 GHz radar demonstrator.

- (d) Your team wants the phased-array demonstrator of part (c) to cover a 10% band. State the conflict this creates with the patch element and one concrete way to resolve it.

Documentation: _____